Geometry-MMALS G1 v1.0.5

Context Mediation, Curriculum Order, and Paired Delta Evidence

Document	Archival protocol report
Scientific status	C0 protocol implementation only
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Primary artifact	Geometry_MMALS_G1_ContextMediation_Curriculum_v1_0_5.ipynb

Purpose

Geometry-MMALS G1 asks whether MMALS learns a functional internal geometry rather than only a geometry of external audit indicators. v1.0.5 narrows the test to context mediation, curriculum robustness, and paired source-level treatment effects.

This release defines the controlled experiment and the evidence bundle. It does not claim that C1-C6 have passed.

1. Evidence motivating v1.0.5

• The v1.0.4 seed-0 run corrected initialization and paired-compute confounds.

• The geometry term improved global route-centroid order more than local source-level route order, while adding almost no predictive benefit by itself.

• A 0.10 old-angle classification anchor improved interpolation, final accuracy, forgetting, route alignment, and absolute causal specificity.

• Context order did not improve while route order did, creating a plausible direct sensory bypass through the router input.

• Absolute causal specificity looked strong in the anchor condition even though the signed effect was reversed.

• Only an ascending curriculum was tested.

v1.0.4 contrast	Observed seed-0 delta	Interpretation
Geometry vs paired compute control	route source rho +0.011; route centroid rho about +0.125; interpolation about -0.003	narrow route alignment effect, little utility gain
Anchor vs geometry	interpolation about +0.062; forgetting about -0.035; final accuracy about +0.040	functional retention evidence reorganizes routing

2. Scientific questions and model

Context mediation

Does inferred context materially affect route choice and prediction beyond direct access to z0?

Geometry attribution

Does route geometry improve trained and held-out geometry under matched initialization, source order, forwards, and steps?

Curriculum robustness

Are results stable under ascending, descending, and fixed-permuted angle orders?

Paired evidence

Are source-level treatment-control deltas positive with paired bootstrap confidence intervals?

Signed causality

Does a positive context intervention move routes in the expected increasing-angle direction?

Functional chain under test

image x -> **sensory z0** -> **context c** -> **route r** -> **host synthesis zMM** -> **prediction**

v1.0.5 intervenes specifically on the router inputs. Hosts continue to transform the real sensory representation, preventing a router ablation from becoming a host-input ablation.

3. Experimental variants

Variant	Views	Router	Geo	Anchor	Purpose
current_only_step_matched	current	z0 + c	0	0	step-matched simple reference
paired_compute_matched_no_geometry	seen angles	z0 + c	0	0	paired compute control
paired_geometry	seen angles	z0 + c	0.10	0	geometry attribution
paired_geometry_anchor_010	seen angles	z0 + c	0.10	0.10	retention trade-off
paired_anchor_context_only	seen angles	c only	0.10	0.10	context mediation
paired_anchor_direct_z0	seen angles	z0 only	0.10	0.10	sensory bypass control

4. Context mediation and curriculum controls

Inference mediation interventions

Condition	Router input	Diagnostic meaning
standard	z0 + context	reference
context zeroed	z0 + 0	tests whether context is necessary
context shuffled	z0 + context from another source	tests source-specific context information
z0 router zeroed	0 + context	tests context sufficiency
both zeroed	0 + 0	router prior / bias control

The notebook exports accuracy drop, class log-probability drop, and route shift. Context dependence is interpreted relative to the effect of removing z0 from the router.

Curriculum order

Name	Order	Purpose
ascending	-60, -30, 0, 30, 60	reference order
descending	60, 30, 0, -30, -60	reverse recency bias
permuted	0, -60, 60, -30, 30	non-monotone order sensitivity

5. Evidence partition and paired uncertainty

• **C1 trained geometry:** -60, -30, 0, 30, 60 only.

• **C2 interpolation:** -45, -15, 15, 45 only.

• **Extrapolation:** -75 and 75 reported separately.

• **Uncertainty:** source-level scores first, paired treatment-control deltas second, bootstrap across source IDs third.

• **Prohibited evidence shortcut:** no ordinary p-value over millions of dependent pairwise distances.

6. Signed causal gate and qualification matrix

A high absolute causal specificity ratio is insufficient. A candidate C4 result must pass orientation, monotonicity, identity preservation, and confidence interval requirements simultaneously.

Condition	Required evidence
Specificity	mean CSR exceeds the frozen threshold and lower confidence bound
Orientation	positive scale -> positive route movement; negative scale -> negative movement
Monotonicity	|effect at scale 2| > |effect at scale 1|
Identity	class evidence drop remains below the frozen bound
Replication	source-block uncertainty and multi-seed consistency

Gate	v1.0.5 status before execution	Promotion requirement
C0	protocol implemented	all structural, finite, budget, split, and export assertions
C1	not qualified	frozen paired-delta thresholds and multi-seed lower bounds
C2	not qualified	paired interpolation and route interpolation gains
C3	not qualified	functional locality and cross-seed role matching
C4	not qualified	CSR + orientation + monotonicity + identity + CI
C5	not implemented	reconstructive and synthetic memory transport
C6	not qualified	accuracy-retention-cost advantage over hardened controls

7. Tracked changes

- CHG-105-01 version and commit guard
- CHG-105-02 current-only step-matched control
- CHG-105-03 explicit router mediation modes
- CHG-105-04 inference mediation interventions
- CHG-105-05 C1/C2 evidence separation
- CHG-105-06 paired source-level deltas
- CHG-105-07 paired bootstrap intervals
- CHG-105-08 curriculum controls
- CHG-105-09 signed causal orientation gate
- CHG-105-10 context dependency summary
- CHG-105-11 host evidence retained
- CHG-105-12 clean PDF export
- CHG-105-13 clean release archive
- CHG-105-14 strict epistemic labels
- CHG-105-15 qualification block

8. Archive and execution workflow

Path	Purpose
notebooks/Geometry_MMALS_G1_ContextMediation_Curriculum_v1_0_5.ipynb	primary Colab experiment
docs/reports/Geometry_MMALS_G1_v1_0_5_Context_Mediation_Curriculum_Report.pdf	archival protocol report
docs/reports/Geometry_MMALS_G1_v1_0_5_Context_Mediation_Curriculum_Report.md	editable report source
docs/changes/Geometry_MMALS_G1_v1_0_4_to_v1_0_5.md	tracked change summary
configs/rotated_mnist_g1_v105.yaml	protocol supplement

1. Push and tag the v1.0.5 package, then run with FORCE_REFRESH=True.
2. Execute smoke mode and verify package, split, finite-value, and budget assertions.
3. Execute development mode for one seed and inspect all deltas without promoting claims.
4. Freeze thresholds and weights before pilot qualification.
5. Run five pilot seeds, then ten final seeds, reporting failures.
6. Replicate on a second controlled transformation before any domain-general statement.

Release hygiene

The clean GitHub archive excludes .git, datasets, runtime results, caches, bytecode, editable-install metadata, and local environments. Raw results belong in a separately generated evidence archive tied to the exact commit SHA and manifest.

Qualification boundary

v1.0.5 is ready for smoke and development execution. It is not a result paper and does not establish context mediation, curriculum robustness, causal geometry, host specialization, memory transport, or operational utility.